MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 03 (SOLUTIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If the common difference of an AP is 5, then what is $a_{18} - a_{13}$? [1]

(A) 5 (B) 20 (C) 25 (D) 30

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 25** [0.5 Mark]

We apply the nth term formula $a_n = a + (n-1)d$:

Formula / Theorem Used: $a_p - a_q = (p-q)d$

Justification: [0.5 Mark]

$a_{18} - a_{13} = (a + 17d) - (a + 12d) = 5d = 5 \times 5 = \mathbf{25}$.

Q2. What is the common difference of an AP in which $a_{18} - a_{14} = 32$? [1]

(A) 8 (B) −8 (C) −4 (D) 4

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 8** [0.5 Mark]

Using $a_n = a + (n-1)d$:

Formula / Theorem Used: $a_{18} - a_{14} = (18-14)d = 4d$

Justification: [0.5 Mark]

$$(a + 17d) - (a + 13d) = 32 \implies 4d = 32 \implies d = \frac{32}{4} = \mathbf{8}$$

Q3. Two APs have the same common difference. The first term of one is -1 and that of the other is -8. Then the difference between their 4th terms is: [1]

(A) -1 (B) -8 (C) 7 (D) -9

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(C) 7** [0.5 Mark]

Given: $a_1 = -1$, $b_1 = -8$, and common difference d is equal for both APs.

Formula / Theorem Used: Difference of nth terms with equal d: $a_n - b_n = (a_1 - b_1)$ (independent of n).

Justification: [0.5 Mark]

$$a_4 - b_4 = (-1 + 3d) - (-8 + 3d) = -1 + 3d + 8 - 3d = \mathbf{7}$$

Q4. If 7 times the 7th term of an AP is equal to 11 times its 11th term, then its 18th term will be: [1]

(A) 7 (B) 11 (C) 18 (D) 0

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(D) 0** [0.5 Mark]

We apply $a_n = a + (n-1)d$ to $7a_7 = 11a_{11}$:

Formula / Theorem Used: $7(a + 6d) = 11(a + 10d)$

Justification: [0.5 Mark]

$7a + 42d = 11a + 110d \implies 4a + 68d = 0 \implies 4(a + 17d) = 0 \implies a + 17d = 0 \implies \mathbf{a_{18} = 0}$.

Q5. **Assertion (A):** If the sum of first n terms of an AP is $S_n = 3n^2 + 5n$, then its common difference is 6.

Reason (R): In an AP, the coefficient of n^2 in S_n is equal to $\frac{d}{2}$, which implies $d = 2 \times 3 = 6$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(B) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true; (R) explains (A).** [0.5 Mark]

We expand the standard sum formula as a polynomial in n:

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{d}{2}n^2 + \left(a - \frac{d}{2}\right)n$

Justification: [0.5 Mark]

Verification via terms: $a_1 = S_1 = 3(1)^2 + 5(1) = 8$, $S_2 = 3(4) + 10 = 22 \implies a_2 = S_2 - S_1 = 14 \implies d = 14 - 8 = 6$. Since the coefficient of n^2 in S_n is always $\frac{d}{2}$, $\frac{d}{2} = 3 \implies d = 6$. Both (A) and (R) are true and (R) explains (A).

Q6. Find the 11th term from the end of the AP $10, 7, 4, \ldots, -62$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying Parameters from the Given AP [1 Mark]

Given: First term $a = 10$, common difference $d = 7 - 10 = -3$, last term $l = -62$.
We know the formula for the nth term from the end of a finite AP:

Formula / Theorem Used: n^{th} term from the end $= l - (n-1)d$

Step 2: Computing the 11th Term from the End [1 Mark]

$$a'_{11} = -62 - (11-1)(-3) = -62 - 10(-3) = -62 + 30 = \boxed{-32}$$

Q7. If the sum of first m terms of an AP is equal to the sum of first n terms ($m \neq n$), prove that the sum of its first $(m+n)$ terms is zero. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: In an AP with first term a and common difference d, $S_m = S_n$ where $m \neq n$.
We know the sum formula of k terms of an AP:

Formula / Theorem Used: $S_k = \frac{k}{2}[2a + (k-1)d]$

Step 1: Simplifying $S_m = S_n$ [2 Marks]

$$\frac{m}{2}[2a + (m-1)d] = \frac{n}{2}[2a + (n-1)d]$$

Multiplying both sides by 2 and expanding brackets:

$$2am + m(m-1)d = 2an + n(n-1)d \implies 2a(m-n) + d[(m^2 - n^2) - (m-n)] = 0$$

Factoring $(m-n)$ out using $m^2 - n^2 = (m-n)(m+n)$:

$$(m-n)[2a + (m+n-1)d] = 0$$

Since $m \neq n$ (so $m - n \neq 0$), dividing by $(m-n)$ yields:

$$2a + (m+n-1)d = 0 \qquad \text{— (i)}$$

Step 2: Computing S_{m+n} [1 Mark]

$$S_{m+n} = \frac{m+n}{2}[2a + (m+n-1)d] = \frac{m+n}{2} \times 0 = \boxed{0 \quad \text{[Hence Proved]}}$$

Q8. Solve the equation: $1 + 4 + 7 + 10 + \cdots + x = 287$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: LHS series $1 + 4 + 7 + 10 + \cdots + x$ has constant differences $4 - 1 = 3$, $7 - 4 = 3$, so it is an AP with $a = 1$, $d = 3$, $l = a_n = x$, and $S_n = 287$.
We know the sum and general term formulas of an AP:

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$ and $a_n = a + (n-1)d$

Step 1: Solving for Number of Terms n [2 Marks]

$$\frac{n}{2}[2(1) + (n-1)(3)] = 287 \implies n(2 + 3n - 3) = 574 \implies n(3n-1) = 574 \implies 3n^2 - n - 574 = 0$$

Splitting middle term using $-42 + 41 = -1$ and $(-42)(41) = -1722 = 3(-574)$:

$$3n^2 - 42n + 41n - 574 = 0 \implies 3n(n-14) + 41(n-14) = 0 \implies (n-14)(3n+41) = 0$$

Since number of terms $n \in \mathbb{N}$ $(n > 0)$, $n = -\frac{41}{3}$ is rejected, leaving $n = 14$.

Step 2: Finding $x = a_{14}$ [1 Mark]

$$x = a_{14} = 1 + (14-1) \times 3 = 1 + 13 \times 3 = 1 + 39 = \boxed{40}$$

Q9. If the mth term of an AP is $\frac{1}{n}$ and the nth term is $\frac{1}{m}$, show that the sum of its first mn terms is $\frac{1}{2}(mn + 1)$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: In an AP with first term a and common difference d $(m \neq n)$: $a_m = \frac{1}{n}$ and $a_n = \frac{1}{m}$.
We know $a_k = a + (k-1)d$ and $S_k = \frac{k}{2}[2a + (k-1)d]$:

Formula / Theorem Used: $S_{mn} = \frac{mn}{2}[2a + (mn-1)d]$

Step 1: Solving for d and a [2 Marks]

$a + (m-1)d = \frac{1}{n}$ (i) and $a + (n-1)d = \frac{1}{m}$ (ii). Subtracting (ii) from (i):

$$(m-n)d = \frac{1}{n} - \frac{1}{m} = \frac{m-n}{mn} \implies d = \frac{1}{mn} \quad (\text{since } m \neq n)$$

Substituting $d = \frac{1}{mn}$ into (i):

$$a + \frac{m-1}{mn} = \frac{1}{n} \implies a = \frac{1}{n} - \frac{m-1}{mn} = \frac{m-(m-1)}{mn} = \frac{1}{mn}$$

Step 2: Evaluating S_{mn} [1 Mark]

$$S_{mn} = \frac{mn}{2}\left[\frac{2}{mn} + \frac{mn-1}{mn}\right] = \frac{mn}{2}\left[\frac{2 + mn - 1}{mn}\right] = \boxed{\frac{1}{2}(mn+1) \quad \text{[Hence Proved]}}$$

Q10. CASE STUDY: Athletics Half-Marathon Training Progression

Neha, a disciplined student athlete of Class 10, sets a goal to participate in the National Youth Half-Marathon event (21 km). Under her coach's advice, she designs a structured weekly progressive training schedule to systematically build cardiovascular stamina without risking physical injury. In the first week of her training regimen, Neha runs 3 km every day. At the start of each subsequent week, she increases her daily running distance by a uniform increment of 1.5 km. [4]

(i) Write down the daily running distances completed by Neha during each of the first three weeks. [1]

(ii) In which week of training will her daily running target reach exactly 15 km? [1]

(iii) In which week will her scheduled daily run first reach or exceed the full half-marathon distance of 21 km? [2]

— **OR** —

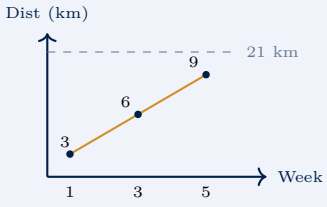
If Neha runs 6 days per week throughout her training, calculate the total cumulative distance she runs over the first 10 weeks. [2]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Extracted Scenario Data: Daily running distances across consecutive weeks form an AP with first term $a = 3$ km and common difference $d = 1.5$ km.

We know the formulas for general term and sum of an AP:

Formula / Theorem
Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$



Sub-part (i): Daily Distances in the First Three Weeks [1 Mark]

Week 1: $a_1 = 3$ km; Week 2: $a_2 = 3 + 1.5 = 4.5$ km; Week 3: $a_3 = 4.5 + 1.5 = 6$ km.

$$\boxed{\textbf{3} \text{ km}, \quad \textbf{4.5} \text{ km}, \quad \textbf{6} \text{ km}}$$

Sub-part (ii): Week When Daily Target Reaches 15 km [1 Mark]

$$3 + (n-1)(1.5) = 15 \implies 1.5(n-1) = 12 \implies n - 1 = \frac{12}{1.5} = 8 \implies n = \boxed{\textbf{9}^{\text{th}} \text{ week}}$$

Sub-part (iii): Week When Daily Run Reaches or Exceeds 21 km [2 Marks]

$$a_n \geq 21 \implies 3 + (n-1)(1.5) \geq 21 \implies 1.5(n-1) \geq 18 \implies n - 1 \geq \frac{18}{1.5} = 12 \implies n \geq \boxed{\textbf{13}^{\text{th}} \text{ week}}$$

Sub-part (iii) [OR Option]: Total Cumulative Distance Over First 10 Weeks [2 Marks]

Sum of 1 daily distance per week across 10 weeks:

$$S_{10} = \frac{10}{2}[2(3) + 9(1.5)] = 5(6 + 13.5) = 5 \times 19.5 = 97.5 \text{ km}$$

Since Neha runs 6 days every week, total distance $= 6 \times 97.5 = \boxed{\textbf{585} \text{ km}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★